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Florian Jaeckle

Lyzeum Ltd.

Dear Florian,

I am writing to apply for the AI Scientist – AI for Medical Imaging position at Lyzeum Ltd. Over 4 years I have designed and shipped computer-vision systems for safety-critical environments, developed expert-level proficiency in Python and PyTorch. I am eager to bring this background to Lyzeum and help expand your interpretable biopsy-analysis models to additional small-intestinal conditions.

At Esbaar I managed a 3.6-million-sample sensor and imagery repository, introduced reproducible experiment tracking with MLflow and DVC, and used Optuna-driven hyper-parameter searches and systematic ablation studies to raise F1-scores on MultiRocket time-series and YOLOv5 models by up to 39 percent. The same data-rigorous discipline will transfer seamlessly to Lyzeum's unparalleled biopsy archive.

While leading development of ViCAS at LivNSense Technologies, I managed a four-engineer team that delivered a vision-based collision-avoidance platform for heavy-industry vehicles. To distinguish forward from reverse motion using the camera feed alone, I designed an optical-flow pipeline paired with a lightweight image-classification network. We then built and deployed object-detection models, optimised them with Intel OpenVINO and mixed-precision quantisation, and reduced inference latency by 72 percent. Throughout, we preserved class-wise heat-map visualisations that frontline safety officers could grasp at a glance—experience directly relevant to clinically interpretable AI.

My projects succeed because I can speak the languages of both technology and domain experts—whether collaborating with rig supervisors in Oman, radiation-protection specialists at the University of Bristol, or, in your setting, pathologists at Addenbrooke's Hospital. I thrive in multidisciplinary teams and am comfortable working on-site several days each week.

Lyzeum's mission—pairing transparent, interpretable models with the world's largest digitised small-intestinal biopsy collection—perfectly matches my conviction that AI must be trustworthy to succeed in high-stakes clinical contexts. Helping to accelerate standardised, rapid diagnosis of coeliac disease and other enteropathies is an opportunity I am genuinely passionate about.

I would welcome the chance to discuss how my experience in vision-model design, dataset stewardship, and cross-functional collaboration can support Lyzeum's path toward clinical trials and regulatory approval. Thank you for considering my application; I look forward to speaking with you soon.

Sincerely,

Dexter Fernandes